

Suricata Network Intrusion Detection System

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In this LAB, I have installed **Suricata IDS** on **Ubuntu server** to detect and visualize network-based threats in real time. This is a step-by-step comprehensive and beginner friendly guide.

Installing Suricata IDS on Ubuntu Server:

The goal of setting up a Suricata home-lab is to gain practical experience in deploying and configuring an Intrusion Detection System (IDS) for network security monitoring. Suricata is an open-source IDS capable of detecting and preventing various network-based threats. This home-lab provides individuals with hands-on experience in setting up, configuring, and utilizing Suricata to enhance network security.

What we'll do?

We will set up a **Suricata IDS** and install it on our **Ubuntu** server. Then we will launch some network probes or network-based attacks to see if Suricata IDS is able to detect that or not.

Steps to configure & install Suricata:

You can find complete commands and guide here:

<https://github.com/0xrajneesh/Suricata-IDS-Home-Lab/blob/main/installing-suricata.md>

1. Install Suricata on the Ubuntu endpoint.

sudo add-apt-repository ppa:oisf/suricata-stable

sudo apt-get update

sudo apt-get install suricata -y

```
root@zayan:/home/zayan# sudo add-apt-repository ppa:oisf/suricata-stable
sudo apt-get update
sudo apt-get install suricata -y
```

- We can confirm if it is installed or not by navigating to **/etc/suricata**

```
root@zayan:/home/zayan# cd /etc/suricata/
root@zayan:/etc/suricata# ls
classification.config  reference.config  suricata.yaml  threshold.config
```

2. Download and extract the Emerging Threats Suricata ruleset:

cd /tmp/ && curl -LO https://rules.emergingthreats.net/open/suricata-6.0.8/emerging.rules.tar.gz

sudo tar -xvzf emerging.rules.tar.gz && sudo mv rules/*.rules /etc/suricata/rules/

sudo chmod 640 /etc/suricata/rules/*.rules

```

root@zayan:/etc/suricata# mkdir rules
root@zayan:/etc/suricata# cd /tmp/ && curl -LO https://rules.emergingthreats.net/open/suricata-6.0.8/emerging.rules.tar.gz
sudo tar -xvzf emerging.rules.tar.gz && sudo mv rules/*.rules /etc/suricata/rules/
sudo chmod 640 /etc/suricata/rules/*.rules

```

- After installing the rules, we can confirm if they are installed or not

```

root@zayan:/tmp# cd /etc/suricata/rules/
root@zayan:/etc/suricata/rules# ls
botcc.portgrouped.rules      emerging-current_events.rules  emerging-imap.rules          emerging-pop3.rules          emerging-user_agents.rules
botcc.rules                  emerging-deleted.rules         emerging-inappropriate.rules emerging-retired.rules       emerging-voip.rules
ciarmy.rules                  emerging-dns.rules             emerging-info.rules           emerging-rpc.rules           emerging-web_client.rules
compromised.rules            emerging-dos.rules             emerging-ja3.rules            emerging-scada.rules         emerging-web_server.rules
drop.rules                    emerging-exploit_kit.rules      emerging-malware.rules        emerging-scan.rules          emerging-web_specific_apps.rules
dshield.rules                 emerging-exploit.rules          emerging-misc.rules           emerging-shellcode.rules     emerging-worm.rules
emerging-activex.rules        emerging-ftp.rules              emerging-mobile_malware.rules emerging-smtp.rules           threatview_CS_c2.rules
emerging-adware_pup.rules     emerging-games.rules            emerging-netbios.rules        emerging-snmpp.rules         tor.rules
emerging-attack_response.rules emerging-hunting.rules           emerging-p2p.rules            emerging-sql.rules
emerging-chat.rules           emerging-icmp_info.rules        emerging-phishing.rules       emerging-telnet.rules
emerging-coinminer.rules      emerging-icmp.rules             emerging-policy.rules          emerging-tftp.rules

```

3. Modify Suricata settings in the `/etc/suricata/suricata.yaml` file and set the following variables:

```

GNU nano 7.2 /etc/suricata/suricata.yaml
YAML 1.1

```

HOME_NET: "[your ubuntu machine IP]"

EXTERNAL_NET: "any"

```

vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.0.0/16,10.0.0.0/8,172.16.0.0/12]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

```

Before

```

vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.23.131]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

    EXTERNAL_NET: "!$HOME_NET"
    #EXTERNAL_NET: "any"

```

After

```
#EXTERNAL_NET: "!$HOME_NET"  
EXTERNAL_NET: "any"
```

After

default-rule-path: **/etc/suricata/rules**

rule-files:

- **"*.rules"**

```
default-rule-path: /var/lib/suricata/rules  
rule-files:  
- suricata.rules
```

Before

```
default-rule-path: /etc/suricata/rules  
rule-files:  
- "*.rules"
```

After

Global stats configuration

stats:

enabled: Yes

```
# Global stats configuration  
stats:  
  enabled: yes
```

Linux high speed capture support

af-packet:

- interface: ens33 (you can check your network interface by running **ip a** or **ifconfig** command)

```
# Linux high speed capture support  
af-packet:  
- interface: eth0
```

Before

```
# Linux high speed capture support
af-packet:
- interface: ens33
```

After

- Save the file
4. Restart the Suricata service:

sudo systemctl restart suricata

```
root@zayan:/etc/suricata/rules# systemctl restart suricata
root@zayan:/etc/suricata/rules# systemctl status suricata
● suricata.service - Suricata IDS/IPS/NOM/FW daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-08-23 14:41:00 UTC; 21s ago
     Docs: man:suricata(8)
           man:suricata-sc(8)
           https://suricata.io/documentation/
  Process: 12483 ExecStartPre=/bin/rm -f /run/suricata.pid (code=exited, status=0/SUCCESS)
 Main PID: 12485 (Suricata-Main)
    Tasks: 8 (limit: 2210)
  Memory: 35.8M (peak: 36.0M)
     CPU: 583ms
    CGroup: /system.slice/suricata.service
            └─12485 /usr/bin/suricata --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid --user suricata --group suricata
```

- Now, edit the **ossec.conf** which is in **/var/ossec/etc/ossec.conf**

```
root@zayan:/etc/suricata/rules# nano /var/ossec/etc/ossec.conf
```

- Go to the bottom of the file and add this block before **</ossec_config>**

```
<ossec_config>
  <localfile>
    <log_format>journald</log_format>
    <location>journald</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/ossec/logs/active-responses.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dpkg.log</location>
  </localfile>

  <localfile>
    <log_format>json</log_format>
    <location>/var/log/suricata/eve.json</location>
  </localfile>
</ossec_config>
```

- Whatever log file we need to monitor, we have to add it here. Save and exit

We have added this block so that **Wazuh** can read Suricata's security alerts from **eve.json**, because it's the file where Suricata stores all structured alert data.

Why this line is needed:

Purpose:

It tells Wazuh Agent → "Hey, also read and analyze this log file from Suricata."

Without this line, Wazuh does not know Suricata's logs exist, so no alerts would be visible in Wazuh.

Why specifically eve.json:

- Suricata produces different log files (like stats.log, fast.log, eve.json etc.).
 - Out of these, eve.json is the main detailed log file that contains all Suricata alerts, events, and metadata in JSON format.
 - Other files (like fast.log) are smaller or only summaries. But eve.json is standard and recommended because Wazuh already knows how to parse Suricata JSON events.
-
- Restart the agent to take effect of this change

```
root@zayan:/etc/suricata/rules# sudo systemctl restart wazuh-agent
```

To see logs in real time in CLI, we can use this command:

tail -f filename

```
root@zayan:/etc/suricata/rules# cd /var/log/suricata/
root@zayan:/var/log/suricata# ls
certs  core  eve.json  fast.log  files  stats.log  suricata.log
root@zayan:/var/log/suricata# tail -f suricata.log
```

Nmap Scan:

Now, we will send some probes to our Ubuntu machine and see if we can see any alerts on Wazuh dashboard.

- Go to your kali machine
- Initiate **Nmap scan** (use your ubuntu machine IP here)

```
(root@kali)-[/home/kali]
# nmap -sV 192.168.23.131
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-27 19:57 EDT
Nmap scan report for 192.168.23.131
Host is up (0.00030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.13 (Ubuntu Linux; protocol 2.0)
MAC Address: 00:0C:29:0A:61:78 (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.10 seconds
```

- Now go to the **Wazuh dashboard > Ubuntu agent > Threat Hunting > Events**

W.

Threat Hunting

ubuntu-agent

Aug 27, 2025 @ 04:57:50.116 - Aug 28, 2025 @ 04:57:50.116

1 fields sorted

Full screen

Export Formatted

747 available fields

Columns

Density

timestamp	agent.name	rule.description	rule.level	rule.id
Aug 28, 2025 @ 04:57:31.8...	ubuntu-agent	Suricata: Alert - ET SCAN Suspicious inbound to Oracle SQL port 1521	3	86601
Aug 28, 2025 @ 04:57:31.8...	ubuntu-agent	Suricata: Alert - ET SCAN Potential VNC Scan 5800-5820	3	86601
Aug 28, 2025 @ 04:57:31.8...	ubuntu-agent	Suricata: Alert - ET SCAN Suspicious inbound to MSSQL port 1433	3	86601
Aug 28, 2025 @ 04:57:31.7...	ubuntu-agent	Suricata: Alert - ET SCAN Suspicious inbound to MySQL port 3306	3	86601
Aug 28, 2025 @ 04:57:31.7...	ubuntu-agent	Suricata: Alert - ET SCAN Suspicious inbound to PostgreSQL port 5432	3	86601
Aug 28, 2025 @ 04:56:13.8...	ubuntu-agent	Suricata: Alert - ET POLICY Possible Kali Linux hostname in DHCP Request Packet	3	86601
Aug 28, 2025 @ 04:54:03.8...	ubuntu-agent	PAM: Login session opened.	3	5501
Aug 28, 2025 @ 04:54:03.8...	ubuntu-agent	PAM: Login session opened.	3	5501
Aug 28, 2025 @ 04:54:03.8...	ubuntu-agent	Successful sudo to ROOT executed.	3	5402
Aug 28, 2025 @ 04:53:53.9...	ubuntu-agent	Listened ports status (netstat) changed (new port opened or closed).	7	533
Aug 28, 2025 @ 04:53:49.8...	ubuntu-agent	Wazuh agent started.	3	503
Aug 28, 2025 @ 04:45:29.7...	ubuntu-agent	Apparmor DENIED miknod operation.	4	52004
Aug 28, 2025 @ 04:45:29.7...	ubuntu-agent	Apparmor DENIED miknod operation.	4	52004
Aug 28, 2025 @ 04:45:29.7...	ubuntu-agent	Apparmor DENIED miknod operation.	4	52004
Aug 28, 2025 @ 04:45:29.7...	ubuntu-agent	Apparmor DENIED miknod operation.	4	52004

Document Details

View surrounding documents

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Table JSON

t_index	wazuh-alerts-4.x-2025.08.27
t_agent.id	004
t_agent.ip	192.168.23.131
t_agent.name	ubuntu-agent
t_data.alert.action	allowed
t_data.alert.category	Potentially Bad Traffic
t_data.alert.gid	1
t_data.alert.metadata.confidence	Medium
t_data.alert.metadata.created_at	2010_07_30
t_data.alert.metadata.signature_severity	Informational
t_data.alert.metadata.updated_at	2019_07_26
t_data.alert.rev	3
t_data.alert.severity	2
t_data.alert.signature	ET SCAN Suspicious inbound to Oracle SQL port 1521
t_data.alert.signature_id	2010936

Document Details

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t	data.alert.signature_id	2010936
t	data.dest_ip	192.168.23.131
t	data.dest_port	1521
t	data.direction	to_server
t	data.event_type	alert
t	data.flow.bytes_toclient	0
t	data.flow.bytes_toserver	60
t	data.flow.dest_ip	192.168.23.131
t	data.flow.dest_port	1521
t	data.flow.pkts_toclient	0
t	data.flow.pkts_toserver	1
t	data.flow.src_ip	192.168.23.133
t	data.flow.src_port	42133
t	data.flow.start	2025-08-27T23:57:30.919890+0000
t	data.flow_id	573201114044277.000000
t	data.in_iface	ens33
t	data.ip_v	4

t	data.src_port	42133
📅	data.timestamp	Aug 28, 2025 @ 04:57:30.919
t	decoder.name	json
t	id	1756339051.25775
t	input.type	log
t	location	/var/log/suricata/eve.json
t	manager.name	wazuh-server
t	rule.description	Suricata: Alert - ET SCAN Suspicious inbound to Oracle SQL port 1521
#	rule.firedtimes	5
t	rule.groups	ids, suricata
t	rule.id	86601
#	rule.level	3
🔕	rule.mail	false
📅	timestamp	Aug 28, 2025 @ 04:57:31.804

Information

ID
86601**Level**
3**File**
0475-suricata_rules.xml**Path**
ruleset/rules**Groups**
ids, suricata

Details

If_sid
86600**Event_type**
pattern: ^alert\$**Options**
no_full_log

Related rules

ID ↑	Description	Groups	Compliance	Level	File
86600	Suricata messages.	ids, suricata		0	0475-suricata_rules.xml
86601	Suricata: Alert - alert.signature	ids, suricata		3	0475-suricata_rules.xml
86602	Suricata: HTTP.	ids, suricata		0	0475-suricata_rules.xml

We can see all the alerts here like:

Source IP, Agent IP, Rule Id, Rule Group, which ports were scanned, location of the logs and many more.

We can also these logs in our Ubuntu machine by using the **tail -f eve.json** command

```
cp_flags ts: "02", "tcp_flags tc": "14", "syn": true, "rst": true, "ack": true, "state": "closed", "ts_max_regions": 1, "tc_max_regions": 1}}
{"timestamp": "2025-08-23T15:47:00.476008+0000", "flow_id": "2224273787264598", "in_iface": "ens33", "event_type": "flow", "src_ip": "192.168.23.133", "src_port": 61317, "d
est_ip": "192.168.23.131", "dest_port": 10001, "ip_v": 4, "proto": "TCP", "flow": {"pkts_toserver": 1, "pkts_toclient": 1, "bytes_toserver": 60, "bytes_toclient": 54, "start":
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025-08-23T15:45:59.899874+0000", "end": "2025-08-23T15:45:59.899903+0000", "age": 0, "state": "closed", "reason": "timeout", "alerted": false, "tcp": {"tcp_flags": "16", "
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{"timestamp": "2025-08-23T15:47:00.476063+0000", "flow_id": "2070664300273472", "in_iface": "ens33", "event_type": "flow", "src_ip": "192.168.23.133", "src_port": 61317, "d
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"2025-08-23T15:45:59.912344+0000", "end": "2025-08-23T15:45:59.912353+0000", "age": 0, "state": "closed", "reason": "timeout", "alerted": false, "tcp": {"tcp_flags": "16",
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est_ip": "192.168.23.131", "dest_port": 911, "ip_v": 4, "proto": "TCP", "flow": {"pkts_toserver": 1, "pkts_toclient": 1, "bytes_toserver": 60, "bytes_toclient": 54, "start": "2
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est_ip": "192.168.23.131", "dest_port": 7999, "ip_v": 4, "proto": "TCP", "flow": {"pkts_toserver": 1, "pkts_toclient": 1, "bytes_toserver": 60, "bytes_toclient": 54, "start":
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{"timestamp": "2025-08-23T15:47:00.476164+0000", "flow_id": "1975394595643427", "in_iface": "ens33", "event_type": "flow", "src_ip": "192.168.23.133", "src_port": 61317, "d
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{"timestamp": "2025-08-23T15:47:00.476218+0000", "flow_id": "2001578199425379", "in_iface": "ens33", "event_type": "flow", "src_ip": "192.168.23.133", "src_port": 61317, "d
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"2025-08-23T15:45:59.924780+0000", "end": "2025-08-23T15:45:59.924956+0000", "age": 0, "state": "closed", "reason": "timeout", "alerted": false, "tcp": {"tcp_flags": "16",
```